



Instacart Predictive Analysis: Neural Network Implementation and Scenario Simulation

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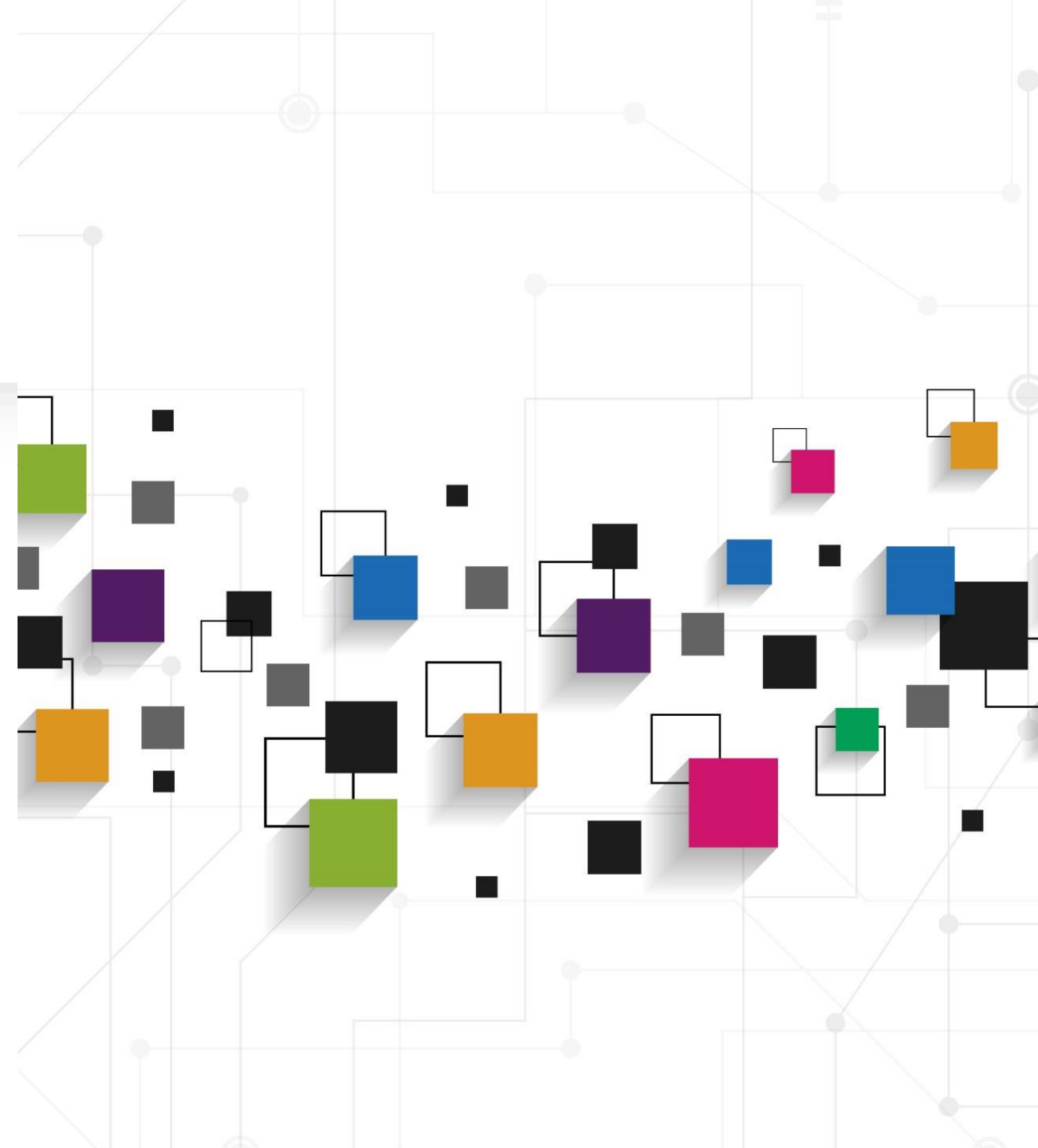
June 13, 2026

Problem Statement

Business Problem: The challenge of accurately predicting whether a grocery item is a routine repeat purchase or a new purchase for Instacart shoppers.

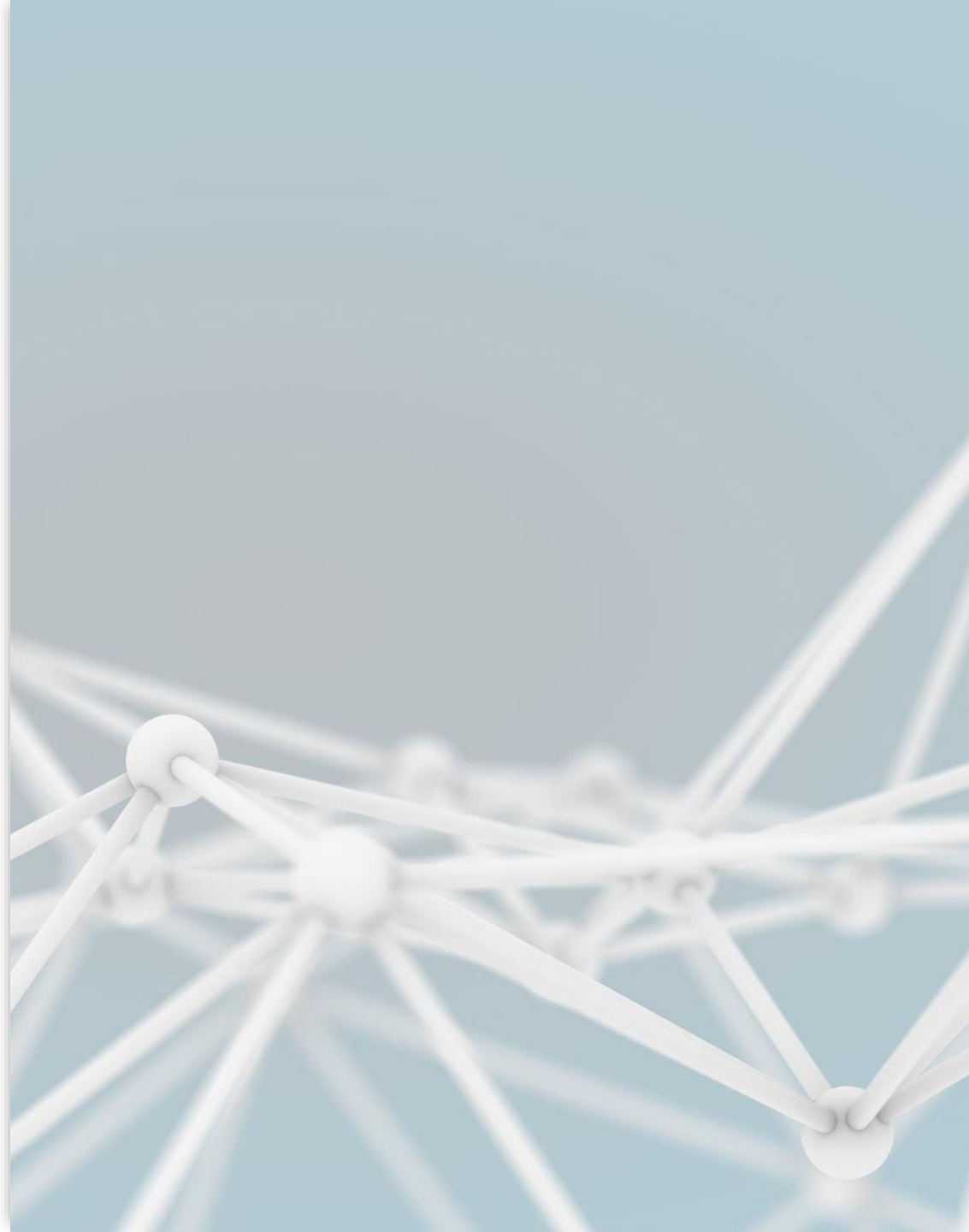
Objective: To improve prediction accuracy by moving from a simple linear model to a more complex nonlinear Multi Layer Perceptron (MLP) Neural Network.

Strategic Goal: Providing actionable data for inventory management, personalized marketing, and automated restock services.



Dataset Overview

- **Dataset:** Instacart Market Basket Analysis (Kaggle).
- **Data Volume:** Over 1.5 million transaction records.
- **Challenge:** Significant class imbalance where routine reorders dominate the dataset.
- **Significance:** This data captures complex human shopping habits that require a deep learning to resolve.

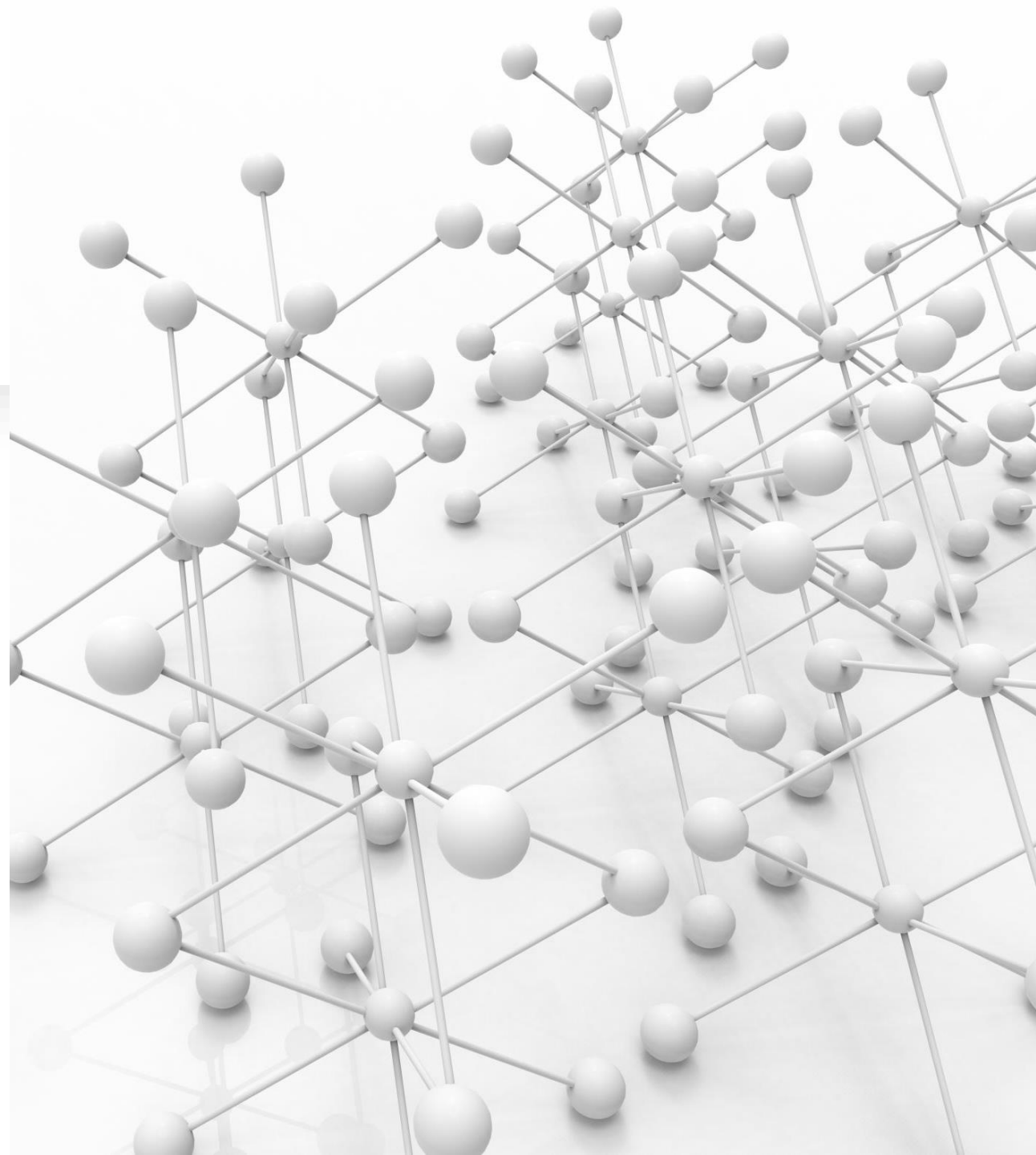


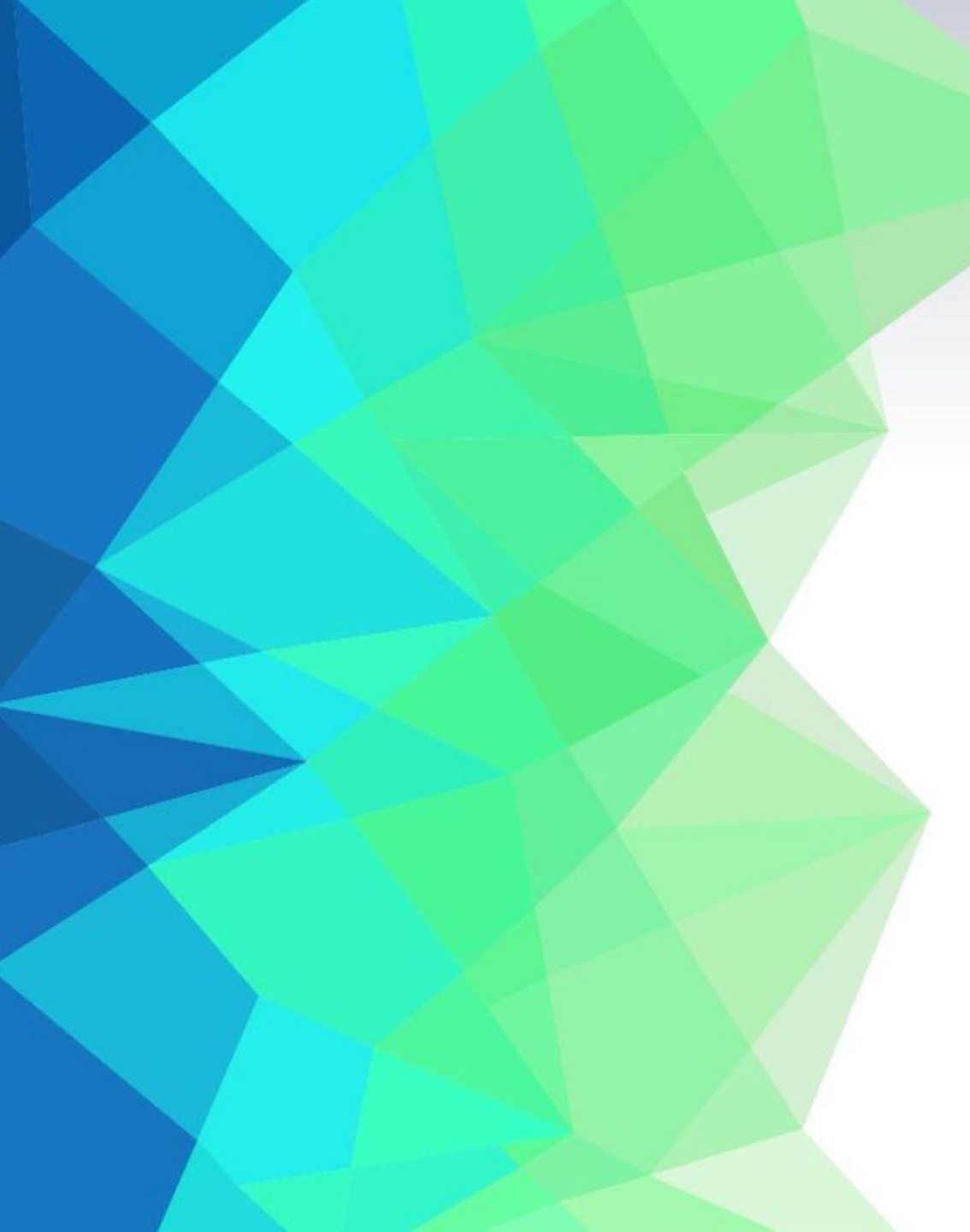
EDA & Preprocessing

- **Preprocessing:** Cleaning, categorical encoding, and feature scaling to prepare data for neural network training.
- **Feature Engineering:** Derived temporal variables ("binned hour of day") to identify shopping patterns.
- **Model Validation:** 80/20 train/test split, ensuring the model was tested against 315,793 unseen transactions.
- **EDA Insight:** Preliminary analysis revealed distinct correlations between shopping frequency, departments, and time of day.

Baseline Model Statistics

- **Model:** Logistic Regression (Linear Benchmark)
- **Accuracy:** 69.14%
- **Performance Metrics:** Precision: 0.68, Recall (Class 0) 0.54, F1-Score: 0.59
- **Key Finding:** The baseline model struggled with class differentiation, indicating a clear need for a more sophisticated architectural approach.





Neural Network

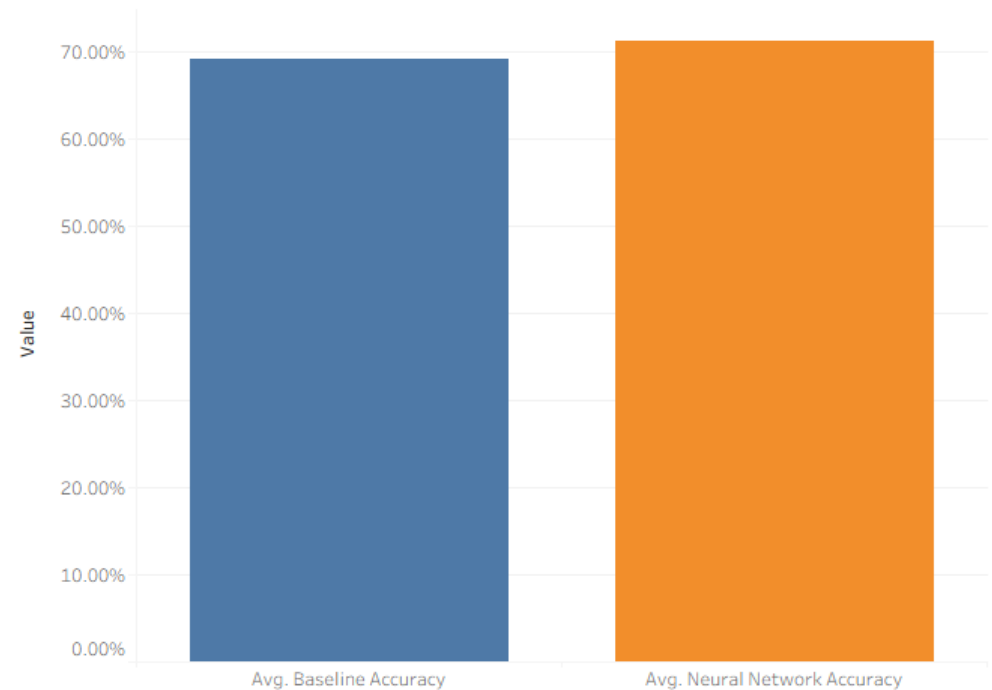
- **Architecture:** Multi Layer Perceptron with two hidden layers (32 and 16 neurons).
- **Activation & Optimizer:** ReLU activation function for internal processing and the Adam optimizer for weight tuning.
- **Accuracy:** 71.29% (2.15% improvement over baseline).
- **Confusion Matrix Insight:** Improved Class 1 recall to 0.84, demonstrating superior detection of routine consumer loyalty.

Baseline vs After Model Comparison

Key Insight: The neural network effectively manages the nonlinear trade offs between precision and recall.

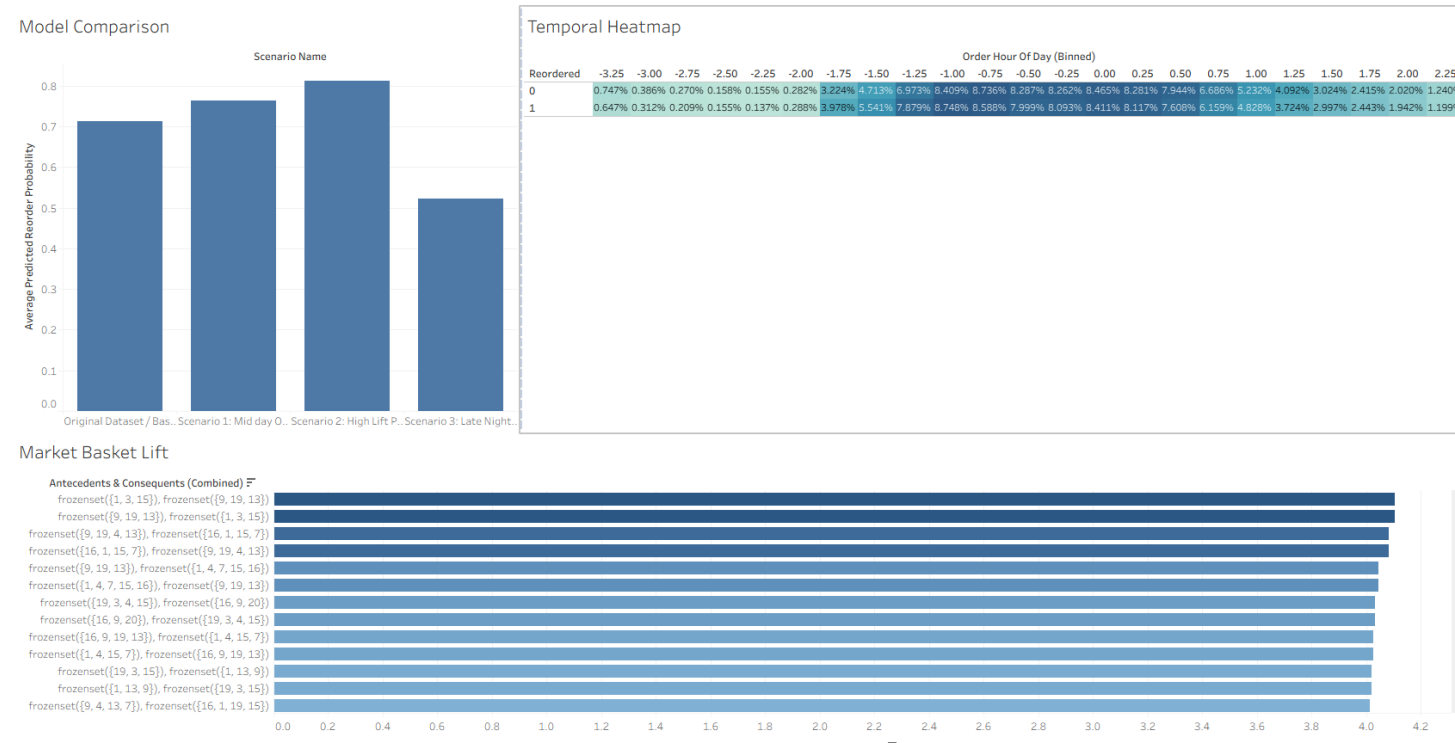
Business Impact: This improvement represents thousands of additional accurate recommendations, directly driving personalization and cart conversion.

Model Comparison



Dashboard Overview

•**Functionality:** An interactive tool allowing stakeholders to visualize reorder probability, temporal trends, and product affinity in real time.



Findings and Recommendations

- Key Finding: Scenario 2 (Promo Bundles) resulted in a peak reorder probability of 81.15%.
- Recommendation 1: Implement targeted cross promotions for "Meal Prep" bundles (Pasta, Canned Goods, Pantry Staples) which show a 4.10 lift score.
- Recommendation 2: Optimize advertising budgets for the 11:00 AM – 5:00 PM window to align with peak reorder density.

Conclusion:

This project successfully delivered an optimized neural network that outperforms linear benchmarks, achieving 71.29% predictive accuracy. By integrating advanced machine learning with interactive dashboard simulation, I have created a scalable and data driven framework that effectively bridges the gap between raw consumer transaction patterns and actionable retail strategy.

References

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